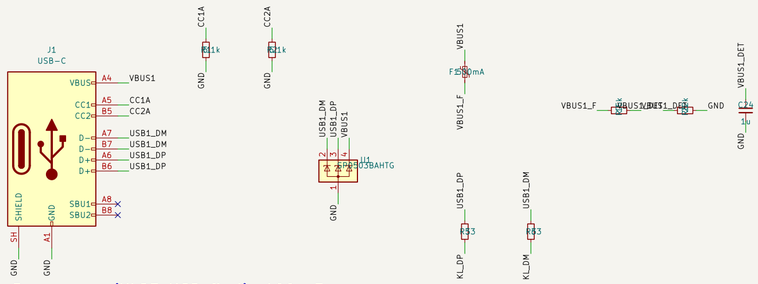
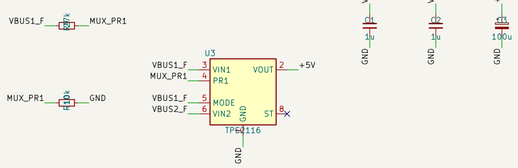
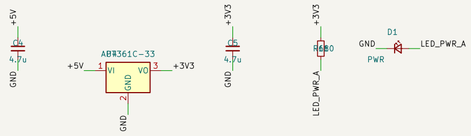
KL25 USB-C, ESD, fuse, VBUS detect



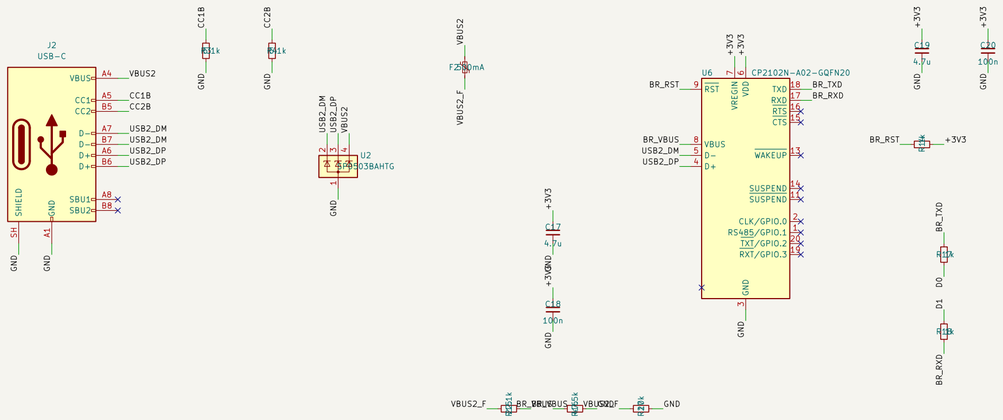
Power mux (KL25 USB first), 100 uF



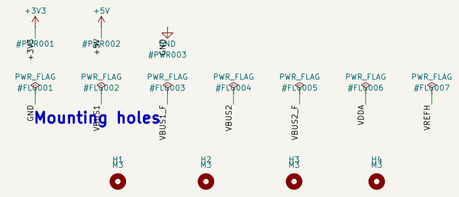
3.3 V LDO, power LED



UART USB-C and CP2102N (on 3.3 V, VBUS sensed)

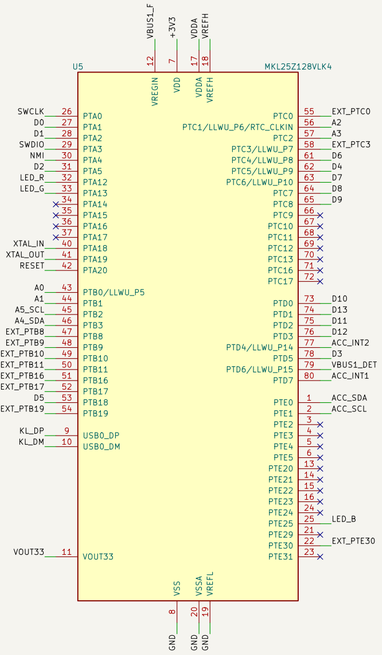


Power nets

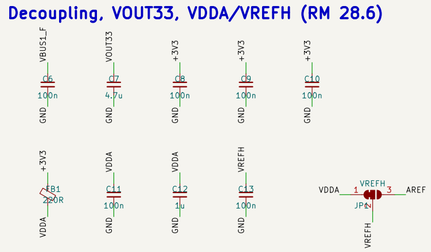


Mounting holes

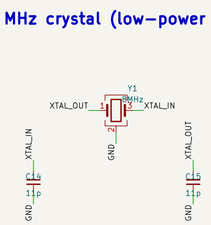
MKL25Z128VLK4



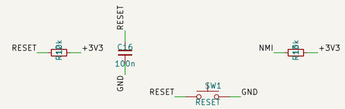
Decoupling, VOUT33, VDDA/VREFH (RM 28.6)



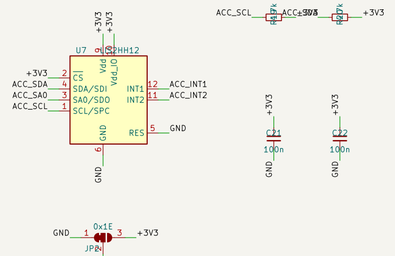
8 MHz crystal (low-power mode)



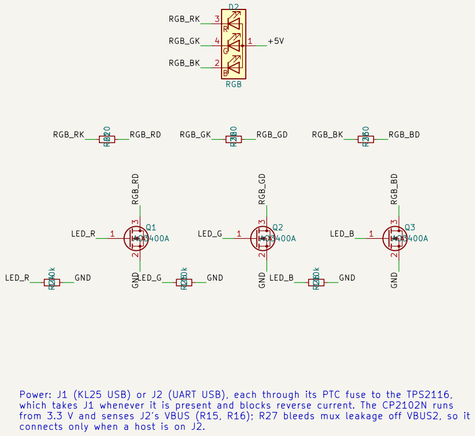
Reset and NMI (KLQRUG 2.1.3.4.1)



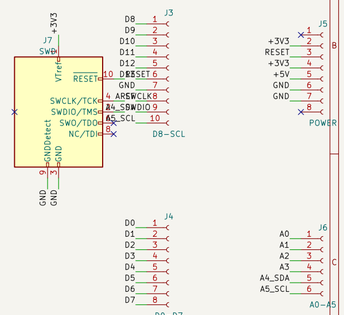
Accelerometer, I2C1



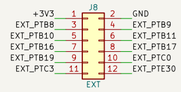
RGB LED, switched from 5 V



Uno R3 shield headers



Extra I/O



Power: J1 (KL25 USB) or J2 (UART USB), each through its PTC fuse to the TPS2116, which takes J1 whenever it is present and blocks reverse current. The CP2102N runs from 3.3 V and senses J2's VBUS (R15, R16); R27 bleeds mux leakage off VBUS2, so it connects only when a host is on J2.

EoS KL25 Board, rev A. Designed by Kartikeydheer Srivastava. Fits Uno R3 shields; 3.3 V I/O.
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Values from NXP KL25P80M48SF0 (data sheet), KL25P80M48SF0RM, KLQRUG, AN3208, AN1706; Silicon Labs CP2102N Rev. 1.5, TI SLVSFG1A (TPS2116), Diodes DS37274 (AP7361C), ST LIS2HH12.
Firmware: run the 8 MHz oscillator in low-power mode (MCG_C2[HGO] = 0): high-gain mode would overdrive the crystal (AN3208 Eqn. 1), and RF must not be fitted (DS Table 19).

Uno R3 mounting holes (4 x 3.20 mm at 15.24 / 2.54 / 50.80 / 66.04 / 17.78 / 45.72 mm) and p
Uno R3 outline (2700 x 2100 mil) and shield header positions: Matthew Beckler, "Arduino Uno R3" di
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Kartikeydheer Srivastava		
Sheet: /		
File: eos-kl25-board.kicad_sch		
Title: EoS KL25 Board		
Size: A2	Date: 2026-10-12	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1